ABHAY PRATAP SINGH (Software Development / Cybersecurity / AI-ML)

Scientist 'B' @C-DOT (DoT, GoI) | Research @YorkU | WhizHack | Microsoft CyberEngage'22 | IIT Jodhpur

Work Experience

Centre for Development of Telematics (DoT, GoI) – Scientist 'B' (Software Developer and Architect) _Jun'24 - Present_

National Scale Packet Processing Solution

- **Designed and developed** high-performance **DPDK** based **packet processing application (C based)** in single-box solution sustaining **400 Gbps** Rx/Tx at ISP gateway.
- Designed a **Security Architecture** based on **Defence-in-Depth** and **Zero Trust** principles for **National Scale Project** spanning multi-state agencies and sites.
- **Implemented DPI (C based)** to surface **eSNI** within UDP flows by parsing Initial ClientHello and TLS extensions via **nDPI** and integrated a **two-phase stateful** pipeline (a fast-path stateless step, followed by a deep inspection stage), maintaining per-flow state to **correlate packets** and trigger actions with minimal latency.
- Engineered end-to-end **DPI strategy** and **architecture**, expanding application coverage by **200%** leading to PoC acceptance by an **Inter-ministerial** committee (GoI).
- **Optimized packet processing code** to leverage NIC offloads, pre-built packets, and batched Tx; achieving **10× throughput uplift** and **15× reduction** in CPU utilization.

Corporate Security & Threat Intelligence

- **Deployed** DNS Response Policy Zone **(DNS RPZ)** across corporate DNS infrastructure in Delhi & Bangalore, blocking **thousands** of requests to blacklisted domains and known/listed C&C servers improving **corporate security posture**.
- **Designed and developed** highly optimized, large-scale crawlers for **Telegram** and **Reddit** using **Playwright/Python** (**70% improvement** over previous solution) to gather multi-format data for **threat intelligence** on various Darknet marketplaces enhancing cyberspace awareness.

York University (Toronto, Canada) – MITACS Research Intern (AI-ML and Cybersecurity Researcher) _Jun'23 - Aug'23_

Automated Intelligence-Driven Malware Detection and Analysis

- Implemented ***Artificial Intelligence & Machine Learning techniques*** like **information-gain selection** and **three-layer Random Forest** architecture plus **weighted score fusion core** for multi-source behavioural analysis over network flows and memory dump features.
- **Developed** an automated **Python + QEMU testbed** where one Windows 10 VM per sample (8 vCPU/8 GB RAM) is "aged" for one week to defeat sandbox evasion, then 2000 malware samples are executed across 8 families, generating a **17TB database** of pcaps and memory dumps.
- **Performance**: L1 (network binary) macro **P/R/F1 ≈ 0.96/0.96/0.95**; L2 (network family) **macro F1 ≈ 0.98** with ~15% Unknown; L3 (memory family) macro **F1 ≈ 0.98**.

VolMemLyzer-V2

- **Designed and developed** a feature extractor tool – VolMemLyzer - V2 **(Python based)** with **250+** supported Memory Analytics Features (compared to <75 in V1) using **Volatility3** framework to provide in-depth memory forensics capabilities.
- **Implemented** modular pipeline which reduces processing duration of stored memory snapshots by **20–30%**, and producing CSVs for downstream **Machine Learning**.

WhizHack Technologies – Security Consultant (Cybersecurity Intern) _Nov'22 - Dec'22_

- Contributed to the **Operational Technology (OT)** security module by implementing **standards (RFC 5424/5425)** for their next-gen SIEM system, TRACE.
- Designed and deployed **3 customized honeypots** for increased efficacy using **CONPOT**, implemented within **containerized Docker environments**.

Microsoft Cybersecurity Engage 2022 – Mentee (Cybersecurity Intern) _May'22 - Jul'22_

- **Ranked 1st (All India)** on the leaderboard based on performance in the program on defending **Critical Infrastructure** (**Delhi SLDC)** from Stuxnet-like cyber-attacks.
- Conducted **OSINT** and identified 10 most probable **attack vectors**, proposing a comprehensive Solution Architecture with 15 **security controls/policies** to minimize the attack surface and implement Defence-in-Depth/Zero Trust principles, alongside incident response and legal considerations.
- Presented 5 functional prototypes/PoCs, including **Machine Learning model** to detect botnet attacks, building **enumeration tools** from scratch, etc.

Technical Skills, Corporate Trainings & MOOCs

♦ Corporate Trainings • MOOCs

Programming/Scripting	C/C++, Python, Bash, PowerShell, JAVA, MATLAB	**Trainings/ MOOCs**	♦ Hacking Android, iOS, and IoT apps by example by c0c0n ♦ High Perf. Computing & Linear Prog. by IISc Bangalore* • Introduction to Digital Forensics by Security Blue Team • Data & Tools for Defense Analysts by Splunk STEP
Cybersecurity Tools/ Application Security	Volatility, Metasploit, Wireshark, NMap, YARA, BurpSuite, Autopsy, Splunk, Suricata, Frida, Magisk, Xposed, JadX, R2		
AI-ML Techniques	CNN, RNN, Random Forest, XGBoost, SVM, KNN, PCA		
Network Software Development	Data Plane Development Kit, Vector Packet Processor, nDPI, BGP, DNS RPZ, QUIC, TCP, DNS, ICMP, OSI Model	**CI/CD Tools**	Docker, Docker Swarm, Github Workflows, QEMU, VMware Tools integration, Kubernetes, Playwright

Education & Certification

* Under Training

Degree	Institute	Grade	Remarks	Year
B. Tech. (CSE)	**Indian Institute of Technology, Jodhpur**	7.88/10.0	Institute Silver Medal	2020-2024
AISSCE (Class XII)	**Army Public School, Birpur**	93.80%	Top 0.1% nationally	2018-2019
AISSE (Class X)	**Army Public School, Birpur**	10.0/10.0	School Topper	2016-2017

Certifications	Issuing Body	Year
Certified Ethical Hacker v13 (Training Completion Certificate)	EC Council	2026*
CompTIA Security+ (SY0-701)	CompTIA	2025
Certified in Cybersecurity	ISC2	2024
Google Professional Cybersecurity Certificate	Google	2023

*Employer Sponsored (Exam Pending)

Publications

Unveiling Evasive Malware Behavior: Towards Generating a Multi-Sources Benchmark Dataset and Evasive Malware Behavior Profiling Using Network Traffic and Memory Analysis | The Journal of Supercomputing (Springer)

Achievements

- Received **Student Distinguished Services Award** and **Institute Silver Medal** for my contributions to Student Senate IITJ in AY 2022-23.
- **Contingent Leader** of 150 IIT-J students in the 55th INTER IIT SPORTS MEET 2022; personally led Runner-Up awardee **Best Marching Contingent** at IIT Delhi.
- Achieved **All India Rank 1** in Microsoft Cybersecurity Engage 2022.
- **State Rank 3** (Uttarakhand) in NSTSE-2019 conducted by Unified Council.

Position of Responsibilities/Recognition

C-DOT	• Led the core **DPI track** (3 members including interns) of National Scale Packet Processing Solution. • **Highest KRA** points of all **<3 YoE** team members in National Scale Packet Processing Solution Team.
IIT Jodhpur	• **Vice-President of Board of Student Sports** (Highest student position in sporting fraternity of IITJ); headed investments and events worth **2.1 Cr (INR)** successfully with 95% budget utilization expanding from just 9 to 15 societies/clubs, which **tripled** student engagement. • Spearheaded successful events like **VARCHAS** (Inter-College), **KRIDANSH** (Inter-Hostel), etc; leading 100s of students hierarchically. • Bagged **4th position** out of 23 IITs; leading a team of 4 members in INTER IIT TECH MEET 12.0 in Cybersecurity PS at IIT Madras.

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